MIL-A-22145B(AS)
AMENDMENT-3
25 October 1972
SUPERSEDING
AMENDMENT-2
28 FEBRUARY 1970

MILITARY SPECIFICATION

ACCELEROMETER GROUP, COUNTING TYPE
DESIGNATION MS25447 AND MS25448

This amendment forms a part of Military Specification MIL-A-22145B(AS) dated 22 May 1967 and has been approved by the Naval Air Systems Command, Department of the Navy.

Page 1

1.1: Delete and substitute:

"1.1 Scope - This specification covers the design and performance requirements for a single axis, four load-factor level Counting Accelerometer Group."

*Page 1

1.2: Delete "Transducer MS25447 (1 through 10)", and substitute "Transducer MS25447".

Page 5

3.3.4: Delete and substitute the following new paragraph:

"3.3.4 Interchangeability - The equipment shall meet the interchangeability requirements of MIL-E-5400."

Add new paragraph:

"3.3.4.1 Interchangeability of Reordered Equipment - For reordered equipment, interchangeability shall exist between all units of any previously manufactured equipment as supplied or designated by the procuring activity. Such interchangeability shall be measured against the supplied or designated model, drawings, and other technical information provided for this purpose. In the event the contract or order does not stipulate whether the model, drawings, or other information shall govern, the designated model shall be used."

FSC 6610

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Page 5 (Cont'd)

3.3.6: Change first sentence to read:

"The equipment shall be capable of satisfactory operation while using the external wiring conforming with the detailed requirements of Figure 1 and 3.4.1."

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3.4.1: Delete and substitute:

"3.4.1 Counting Accelerometer Group - The equipment shall be utilized to count aircraft maneuver - load factor. The counting accelerometer group shall consist of two components:

(a) A hermetically sealed force-balance accelerometer referred to as the transducer: which shall furnish voltage signal outputs, proportional to applied accelerations, provide filter circuitry for required dynamic response, and test circuits; and

(b) A counting accelerometer display referred to as the indicator: which shall have electromechanical counters with associated circuits to indicate the number of times the aircraft is subjected to load factors of preset levels, a group power supply section with voltage regulation providing 18.5 ± 0.5 VDC for the proper signal voltage functions of both the indicator and the transducer, a delay circuit permitting satisfactory load level counting without interference, an inhibitive circuit precluding registration of improper counts during initial application of power, and a digital lockout circuit which precludes registration of counts when the load factor maneuver has a duration of less than 150 milliseconds.

The accelerometer group, when electrically interconnected, shall provide signals and control circuits as shown in Figure 1.

(1) The transducer shall provide signals on wire numbers 3 and 4 with system reference to wire number 1 as follows:

	V.D.C.	
LOAD	Wire 3	Wire 4
+lg	1.1 ± 0.9	0.9 ± 0.4
Reset level	7.8 ± 2.0	0.9 ± 0.4
Counter #1 level	12.5 ± 1.0	0.9 ± 0.4

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LOAD	V.D.C. Wire 3	V.D.C. Wire 4
Counter #2 level	12.5 ± 1.0	6.8 ± 1.3
Counter #3 level	12.5 ± 1.0	11.45 ± 0.85
Counter #4 level	12.5 ± 1.0	18.0 ± 0.5

(2) The indicator shall be capable of proper respondence to the above signal level of the transducer as specified.

(3) The following equipment impedances shall apply for Figure 1:

J101 Pin Numbers	Equipment Impedance Transducer Ohms	Equipment Impedance Indicator Ohms
1	reference +	reference +
2	225 ± 12	1 Max
3	4600 ± 250	100K ± 10K (a) 40 Max (b)
4	55 ± 10	1-5M
5	1650 ± 50	(c)
6	0 to 20	(c)

All impedance shall be measured with respect to pin 1 J101
+ system ground (common for wires 2, 3 and 4)
(a) when signal voltage is less than 12 VDC
(b) when signal voltage is 12 VDC or greater
(c) not to be used."

3.4.1.1 Delete and substitute:

"3.4.1.1 Wiring - The wiring of the accelerometer group shall be in accordance with MIL-E-5400 and shall operate satisfactorily in the environment stated herein. The wiring of each component shall comply with the requirements specified in 3.3.6, 3.4.1, and Figure 1."

3.4.1.2 Delete and substitute:

"3.4.1.2 Electrical Protection - The equipment shall provide electrical protection for overvoltage and undervoltage in accordance with MIL-E-5400 and shall automatically resume normal operation when the voltages return within limits."

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3.4.1.6: Delete and substitute:

"3.4.1.6 Dynamic Response - The dynamic response of the counting accelerometer group to sinusoidal inputs shall be flat to within ±2% from zero to 2.5 Hz. Above 2.5 Hz, the output per "g" shall decrease smoothly at an average rate not less than 18 db per octave. The equipment shall indicate counts at single-amplitude load-factor levels prescribed in Table 1, within the following limits:

(1) ±0.15g for load-factor levels of 2.0g to 5.0g.

(2) ±0.20g for load-factor levels of 5.5g to 7.0g.

(3) ±0.30g for load-factor levels of 7.5g to 10.0g.

The indicator shall not indicate a count on operation with sinusoidal or square wave inputs of 4.0 Hz and above."

Page 8

3.4.2: Delete and substitute:

"3.4.2 Transducer - The transducer shall have four load-factor levels and a reset level. The levels shall be the values specified in Table 1."

*Page 8

Table 1: Delete and substitute:

"TABLE 1

Load-Factor and Reset Levels

MS Part No.	Counter 1	Counter 2	Counter 3	Counter 4	Reset Levels
MS25447-1	2.00	2.50	3.00	3.50	1.50 + 0 - 0.25
MS25447-2	2.50	3.50	4.00	4.50	1.75 ± 0.20
MS25447-3	2.50	3.50	4.50	5.50	1.75 ± 0.20
MS25447-4	3.00	4.00	5.00	6.00	2.00 ± 0.20
MS25447-5	4.00	4.50	5.00	5.50	2.50 ± 0.20

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*Page 8 (Cont'd)

TABLE 1

MS Part No.	Counter 1	Counter 2	Counter 3	Counter 4	Reset Levels
MS25447-6	4.00	5.00	6.00	7.00	2.50 ± 0.20
MS25447-7	5.00	6.00	7.00	8.00	3.00 ± 0.20
MS25447-8	5.50	6.00	6.50	7.00	3.25 ± 0.20
MS25447-9	6.00	7.00	8.00	9.00	3.50 ± 0.20
MS25447-10	6.00	7.00	8.50	10.00	3.50 ± 0.20
MS25447-11	2.50	3.00	3.50	4.00	1.75 ± 0.20
MS25447-12	2.50	3.50	4.50	5.00	1.75 ± 0.20
MS25447-13	3.00	3.50	4.00	4.50	1.75 ± 0.20
MS25447-14	3.50	4.00	4.50	5.00	2.50 ± 0.20
MS25447-15	3.50	4.50	5.50	6.00	2.50 ± 0.20
MS25447-16	6.50	7.50	8.50	9.50	3.50 ± 0.20
MS25447-17	7.00	8.00	9.00	10.00	3.50 ± 0.20"

3.4.2.3: Delete second sentence.

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3.4.3.5: Second sentence; delete and substitute:

"The digit height shall be 0.150 to 0.201 inch, and the line thickness shall be 0.018 to .027 inch."

3.4.3.6: Delete and substitute:

"3.4.3.6 Markings - Each counter of the indicator shall be identified by number. The lowest number shall correspond to the lowest load level counter until all counters are identified."

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4.2: Line 4; Delete "4" and substitute "4.7".

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4.2.2: Delete and substitute:

"4.2.2 Scope of Tests - Preproduction tests shall include all the tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. As a minimum preproduction tests shall include all the tests of this specification (see 4.2.2.1, 4.2.2.2, 4.4 and 4.5), the interference tests of MIL-I-6181, and the following tests of MIL-T-5422:

(1) Humidity (except that in steps 1 and 2, the temperature 50°C shall be replaced by a temperature of 72°C).

(2) Fungus Resistance

(3) Salt Spray

(4) Explosion Proof"

4.2.2.1: Delete and substitute:

"4.2.2.1 Acceleration Test - With electrical power off, the accelerometer group shall not record any counts when subjected to 10g for one minute along each of three mutually perpendicular axes. (One of these mutually perpendicular axis shall be the sensitive axis of the transducer.) With electrical power on, the accelerometer group shall record only one count when subjected to 10g along the sensitive axis of the transducer with the indicator in its normal position. This test shall be repeated with the transducer remaining in the sensitive axis position and the indicator rotated to each of the remaining mutual perpendicular axis. Upon completion of this test, the equipment shall be subjected to the static calibration test."

4.2.2.2: First sentence; Delete the word "indicator" and substitute "counting accelerometer group".

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4.4.1.4: Sentences 2 and 3; Delete and substitute:

"All calibrations shall remain within ±0.10g of the levels specified in Table 1 for load factors of 2.00 to 7.00g and ±0.15g for load-factor levels of

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7.50 to 10.00g. These limits shall apply before and after the transducer is subjected to any and all environmental tests."

Page 14

4.4.1.7: Delete title and substitute: Counting Rate Test

First sentence; Delete and substitute:

"The counting accelerometer group shall be subjected to a sinusoidal or a square wave input of 1,2 and 2.5 cps."

4.4.2: Sentence two; Replace word "Counter" with the word "Counting".

4.4.2.1: Delete and substitute:

"4.4.2.1 Scope of Tests - Each sample selected shall be subjected to the following tests, conducted in the order listed:

Vibration	4.4.2.2
Temperature Shock Test	4.4.2.3
Leak Test	4.4.1.5
Dynamic Test	4.4.2.4
High Temperature Test	4.4.2.5
Low Temperature Test	4.4.2.6"

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4.4.2.4.1: Delete and substitute:

"4.4.2.4.1 Dynamic calibration - The transducer shall be calibrated against known sinusoidal acceleration inputs of 1, 2 and 2.5 cps. The counting accelerometer group shall count at the single-amplitude, load-factor levels prescribed in Table 1 within the limits specified in 3.4.1.6. The calibrations shall remain within these specified limits before and after the counting accelerometer group has been subjected to any or all the environmental tests. This test shall be performed by subjecting the counting accelerometer group to the stated frequencies of this specification and shall be accomplished upon a suitable mechanical dual centrifuge or by approved electrical acceleration simulations."

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4.4.2.4.2: Sentence three; Delete and substitute:

"The superimposed vibration shall be applied at a rate varying logarithmically or linearly over the range of 5 to 120 cps at the amplitude specified in Table 2 for a duration of 5 minutes for each sweep of the specified range."

(1) Line (2); Delete "nominal".

(2) Line (3); Delete (4.6.3) and substitute (4.4.1.4).

(3) Line (5); Delete "nominal".

Add the following paragraph:

"(6) The static load-factor level may be applied by an electrical signal, but the superimposed vibrations of Table 2 must be applied by actual mechanical vibration of the transducer as a unit."

Table 2: Delete and substitute:

"TABLE 2

Frequency (cps)	Amplitude (inch or "g")
5 to 30	0.10 inch double amplitude
31 to 120	4.5g single amplitude"

4.4.2.6: Paragraph 4.4.2.6: Delete sentence two and substitute:

"At the end of 4 hours, and while still at 125°C, the counting rate test, the indicator circuit test, the static calibration test, and the dynamic tests shall be performed."

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4.5: Delete last sentence and add:

"Quantity of Equipments offered for acceptance	Quantity to be selected for life tests
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof"

4.5.2: Delete and substitute:

"4.5.2 Performance Check - At every 10,000 operations at the completion of the life test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity. This performance check shall include subjecting the equipment to and meeting the requirements of 4.4.1.4 and 4.4.2.4."

Page 25

Delete and substitute the revised Figure 1 of this amendment.

Page 26

Delete and substitute the revised Figure 2 of this amendment.

Page 27

Delete and substitute the revised Figure 3 of this amendment.

NOTE: An asterisk preceeding a page number denotes a change to the specification which did not appear in Amendment 2.

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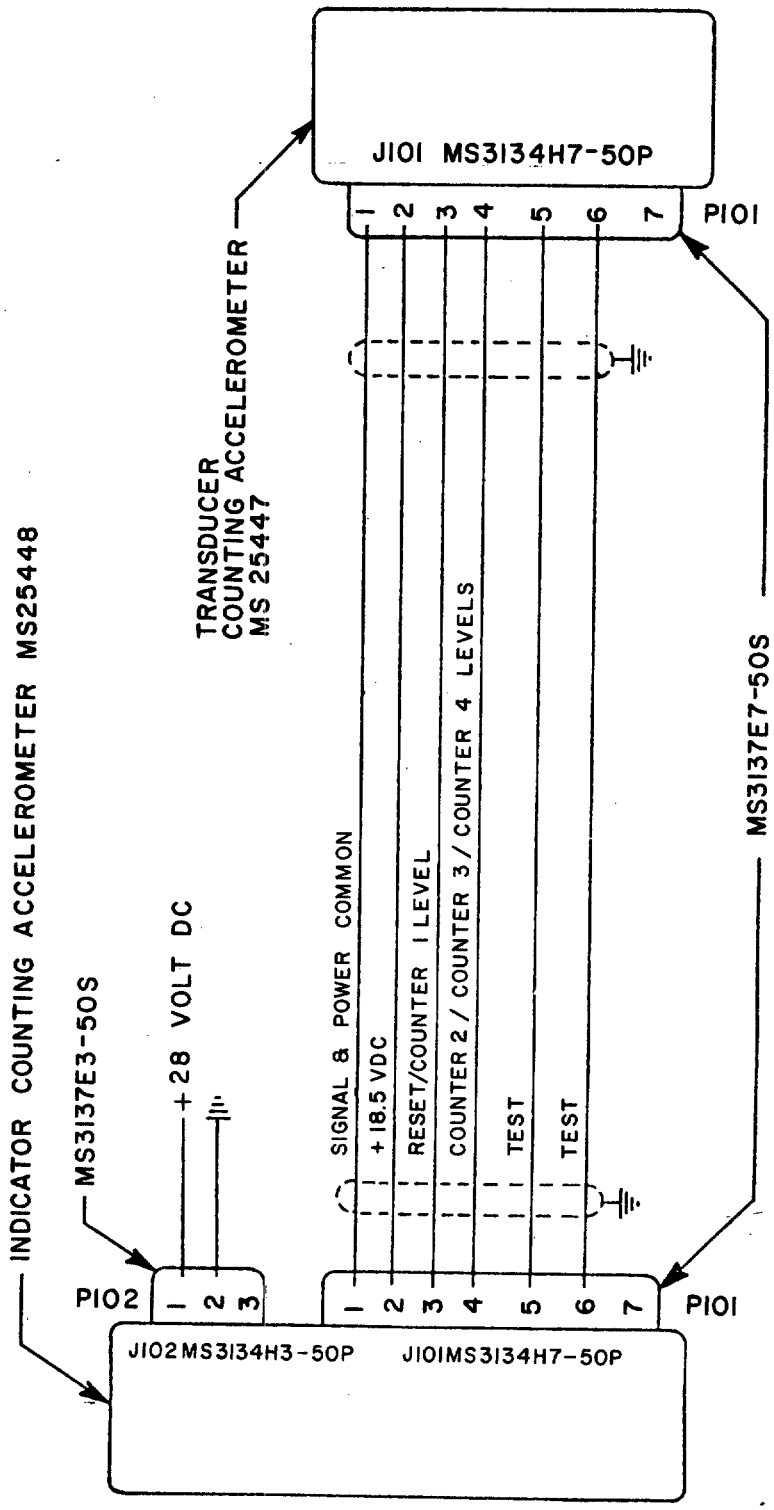
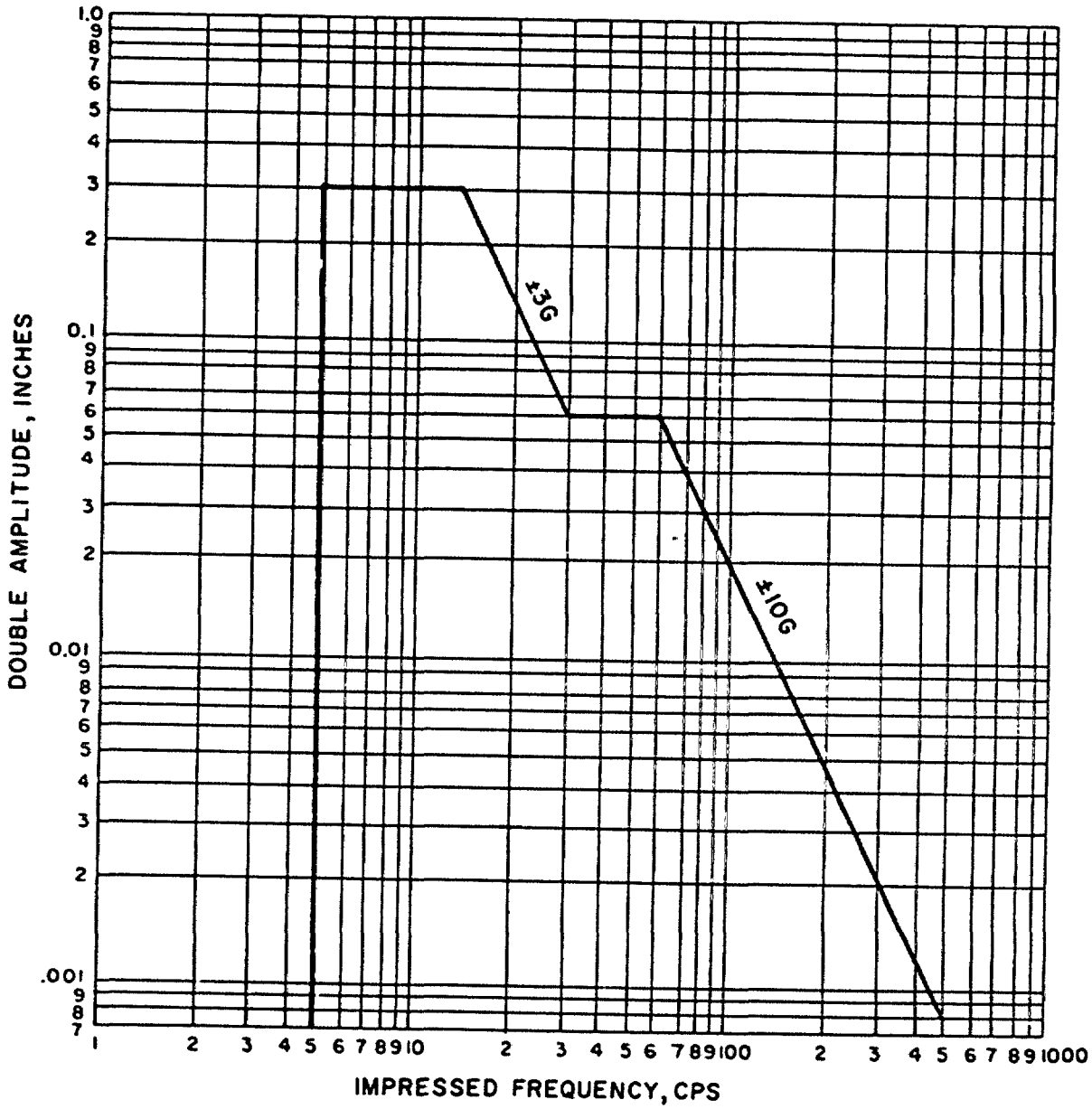


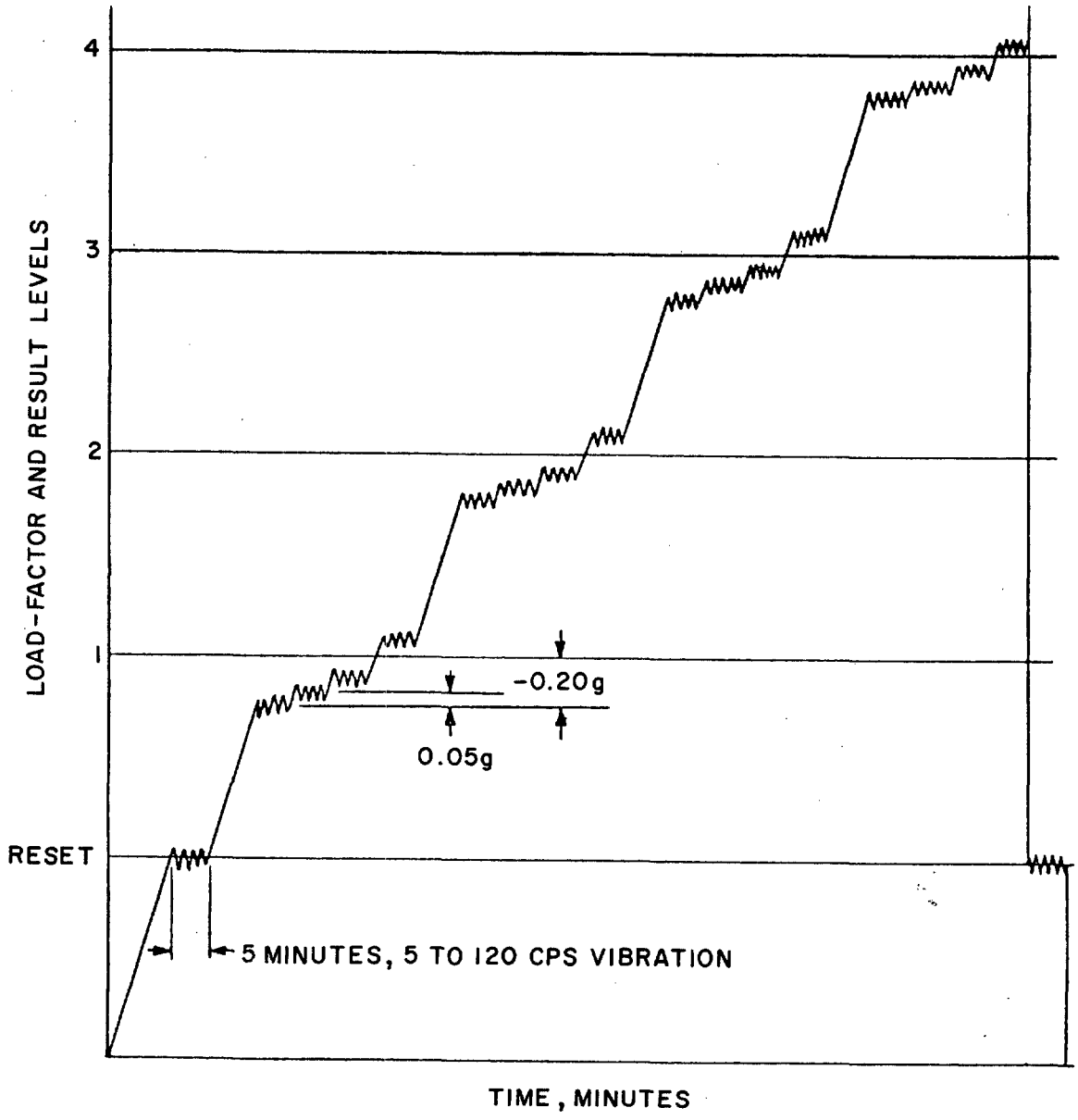
Figure 1

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VIBRATION REQUIREMENTS
Figure 2

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TYPICAL EXAMPLE OF SUPERIMPOSED VIBRATION
APPLIED FOR A TOTAL OF NINETY MINUTES
Figure 3

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